

Chapter # 1 Computer System

Topic 1: Introduction to HCI (Human-Computer Interaction)

- **Ice Breaker (Class shuru karne ka sawal):** Class se puchen, "Jab aap apne mobile par PUBG khelte hain ya WhatsApp par message type karte hain, toh aapko kaise pata chalta hai ke game mein goli chal gayi ya message chala gaya? Aur mobile aapki baat kaise samajhta hai?" (Bachon ka jawab aayega: Touch karne se aur screen dekhne se).
- **Teacher Key Points (Class ko samjhane ke liye):**
 - Aaj hum "Computer Systems" ka pehla chapter parh rahe hain jiska aham topic hai **HCI** yani **Human-Computer Interaction**.
 - Asan lafzon mein: Insaan (Human) aur Machine (Computer/Mobile) aapas mein kaise rabta karte hain, is amal ko HCI kehte hain. Ise CHI ya MMI (Man-Machine Interaction) bhi kaha jata hai.
 - Jaise hum insaan apni 5 hilaayon (senses) ka istemal karte hain, waise hi computer se baat karne ke liye bhi hum yahi senses use karte hain: Sight (Dekhna), Touch (Chhuna), Hearing (Sun-na), Voice (Bolna), aur Spatial (Harkat/Jagah).
- **Real Life Example:** Smartwatch jab aapki heart rate check karti hai aur screen par dikhati hai, toh ye Touch/Spatial aur Sight ka interaction hai.

Topic 2: Types of Interaction (Hum Computer se kaise baat karte hain?)

- **Teacher Key Points:**
 - **Traditional Interaction (Interaction I - Purana Tariqa):** Isme hum aam devices jaise Mouse aur Keyboard (Computer ke liye) ya Touch screen (Smartphone ke liye) use karte hain. Hum button dabate hain aur screen par result dekhte hain.
 - **Natural Interaction (Interaction II - Naya aur Qudrati Tariqa):** Isme computer humari aam zindagi ke gestures (harkaat) aur aawaz samajhta hai, bina keyboard ya mouse ke.
- **Real Life Example:**
 - *Traditional:* Keyboard se MS Word mein type karna.

- *Natural*: Mobile ka "Face Unlock" ya aawaz se Google Assistant/Alexa ko gaana lagane ka kehna (Voice Interaction), ya phir Xbox par hath hila kar bowling game khelna (Body & Gesture Interaction).

Topic 3: Applications of HCI (HCI kahan kahan istemal hota hai?)

- **Teacher Key Points:** HCI har field mein istemal ho raha hai taake software aasan aur user-friendly ban sakay:
 - *Healthcare*: Mareezon ke liye telemedicine apps.
 - *Banking*: Secure aur aasan mobile banking apps jahan fauran paise transfer hote hain.
 - *Education*: E-learning platforms aur exam portals.
 - *Networking*: Social media aur messaging apps (jaise WhatsApp) jahan log aapas mein connect hote hain.

Topic 4: Components of HCI (HCI ke bunyadi hissay)

- **Teacher Key Points:** Ek mukammal HCI system ke 7 aham hissay (components) hote hain:
 1. **User:** Jo system istemal karta hai. Do tarah ke log hote hain: *End Users* (jo directly app chalte hain, jaise hum) aur *Stakeholders* (jo app banwate hain ya jinka isme faida hota hai, jaise company ka boss).
 2. **Computer System:** Wo machine (hardware/software) jis par insaan aur computer aapas mein input/output ke zariye rabta karte hain.
 3. **User Interaction:** Wo tariqa jisse user computer ko hidayat deta hai. Misal ke tor par:
 - *Direct Manipulation*: Screen par icon ko ungli ya mouse se utha kar dusri jagah rakhna (Drag and drop).
 - *Conversational*: Chatbot ya Siri/Alexa se natural language mein baat karna.
 - *Biometric*: Fingerprint ya Iris scan se phone unlock karna.
 4. **Task:** Wo makhsoos kaam jo user karna chahta hai (jaise online ticket book karna ya form fill karna). Har task ka ek 'Time Requirement' hota hai aur 'Complexity' hoti hai.

5. **User Interface:** Wo medium ya rasta jiske zariye hum system se jurte hain. Iski kai types hain:

- *GUI (Graphical User Interface):* Jaise Windows ya Android, jisme tasweerein aur icons hote hain.
- *CLI (Command-Line Interface):* Jaise MS-DOS ya Linux jahan sirf black screen par text (commands) likhna parta hai.
- *Touch, Voice & Natural:* Smartphone ki touch screen, Smart Speakers (Alexa), aur VR (Virtual Reality) headsets.

6. **Environment:** Wo mahol jahan device use ho rahi hai. (Jaise office ki desk, class room, ya bahar sadak par laga ATM).

7. **Feedback:** Jab aap computer ko command dete hain, toh wo jo jawab deta hai. (Jaise file download hote waqt "Loading..." likha aana, ya password galat hone par "Error" aana, ye *System Feedback* hai).

- **Class Activity (Lecture ke aakhir mein):** Bachon ko ek **ATM Machine** imagine karne ko kahein aur unse puchen, "Batao is scenario mein Component 1 se le kar 7 tak kya kya hai?" (Teacher ke liye jawab: User hum hain, Computer ATM machine hai, Interaction touch ya button dabana hai, Task paise nikalna hai, Interface ATM ki screen hai, Environment bank ka ATM room hai, aur Feedback "Transaction Successful" ka message ya slip nikalna hai).

Topic 5: Importance of HCI (HCI kyun zaroori hai?)

- **Teacher Key Points:**

- HCI se machine istemal karna aasan ho jata hai. Agar design acha ho toh kaam jaldi hota hai aur galtiyan kam hoti hain.
- Iske aham fawaid (implications) ye hain:
 1. **Enhance Productivity:** User apna kaam tezi se aur behtar tariqe se kar pata hai.
 2. **Promotes Inclusivity:** Har qisam ke log, chahe unhe dekhne ya sunne ka masla ho, wo bhi technology use kar sakte hain.
 3. **Drives Innovation:** Nayi technologies jaise AR/VR aur AI, HCI ki wajah se hi kamyab hain.

4. **Economic Impact:** User-friendly (aasan) products market mein zyada chalti hain aur log unhe pasand karte hain.
5. **Ethical Responsibility:** Design aesa hona chahiye jo privacy ka khayal rakhe aur sab ke liye faidamand ho.

Topic 6: Accessibility Principles (Sab ke liye aasan design)

- **Teacher Key Points:**

- HCI ka maqsad hai ke system har koi istemal kar sakay, chahay kisi ko dekhne mein ya sunne mein masla ho. Is ke liye kuch asool hain:
 1. **High Contrast:** Likhai aur background ka color aisa ho ke parhne mein aasani ho. *Example:* White background par black text.
 2. **Readable Fonts:** Aise fonts aur size use karna jo aasaani se parhay ja sakein. *Example:* 'Calibri' font.
 3. **Colour-Blind Friendly Design:** Sirf color par depend na karein, labels aur symbols bhi use karein taake colour-blind log bhi samajh sakein.
 4. **Inclusive Navigation:** Sirf mouse nahi, keyboard shortcuts se bhi website/app chalani aasan ho.
 5. **Multimodal Detection:** Information mukhtalif tarikon se di jaye. *Example:* Video ke sath captions (subtitles) takay sunne mein maslay wale log bhi samajh sakein.

Topic 7: Need Analysis (Design se pehle zaroorat samajhna)

- **Teacher Key Points:**

- Koi bhi app ya system banane se pehle yeh jan'na zaroori hai ke users ki kya zaroorat hai. Isay 'Need Analysis' kehte hain.
- Iske steps yeh hain:
 1. **Identify Users:** System kon use karega?
 2. **Information Collection:** Users se interview ya survey karna.
 3. **Analyze User Needs:** Masail aur zarooriyat ki nishan-dahi karna.
 4. **Prioritize Needs:** Konsi zaroorat sab se aham hai.

5. **Develop Solutions:** Un zarooriyat ko pura karne ke liye system design karna.

- **Real Life Example:** Agar bank ki nai app banani hai toh pehle logon se puchenge ke unhe kya features chahiye (jaise fingerprint login ya aasan money transfer).

Topic 8: HCI Problems & Solutions (Masail aur unka hal)

- **Teacher Key Points:**

- **Common Problems:**

- *Interface Complexity:* Screen par bohat zyada options hona jisse samajh na aye ke kya karna hai.
 - *Poor Feedback:* Form submit karne par system kuch na bataye ke kaam hua ya nahi.
 - *Inconsistency:* App ke mukhtalif hisson mein ek hi kaam ke liye alag alag icons hona.
 - *Accessibility Gaps:* Aisa design jo disabled logon ke liye theek na ho.
 - *Contextual Neglect:* App ka raat ke waqt parhne ke liye dark mode na hona.

- **Solutions (Methods to Improve):**

- *Enhance Usability:* Design ko simple aur aasan banana.
 - *Strengthen Accessibility:* Har kisi ke liye kabil-e-istimal banana (jaise captions lagana).
 - *Ensure Consistency:* Pori app mein ek jaisa design aur icons rakhna.
 - *Provide Clear Feedback:* Har action ke baad user ko batana ke kya hua (jaise "Payment Successful").

Topic 9: UI, UX aur Wireframing

- **Teacher Key Points:**

- **UI (User Interface):** Screen par jo kuch nazar aata hai (buttons, menus, text, images). Yeh visual elements hote hain.

- **UX (User Experience):** System istemal karte waqt user ko kaisa mehsoos hua (aasan laga ya mushkil).
- **Wireframing:** App ka asli design banane se pehle uska ek basic sketch (blueprint) banana. Do types hoti hain:
 - *Low-Fidelity:* Sadha black & white sketch (kaghaz par ya simple software par).
 - *High-Fidelity:* Detailed design jo bilkul asli app jaisa dikhta hai.
- **Prototype:** UI ka early version jisko test kiya ja sakta hai ke kya yeh theek se kaam kar raha hai ya nahi.
- **Figma:** Ek web-based tool hai jo UI design karne aur prototypes banane ke liye istemal hota hai.

- **Class Activity:**

Bachon se kaghaz par ek "School Attendance App" ka simple Low-Fidelity wireframe draw karne ko kahein, jisme Login button aur Attendance mark karne ka option ho.

Topic 11: Testing Methods (System ko Test karne ke Tareeqe)

- **Teacher Key Points:**

- Testing ka matlab hai yeh check karna ki banaya gaya product theek se kaam kar raha hai aur users ki zaroorat puri kar raha hai ya nahi.
- Testing ke mukhtalif types yeh hain:
 - **Usability testing:** Users ko app istemal karte hue gaur se dekhna (observe karna) taaki unki efficiency aur satisfaction measure ki ja sake.
 - **Task-based testing:** Users ko ek khaas task dena aur yeh note karna ki unhein kitna time laga, kitni success mili aur kitni galtiyen (errors) hui.
 - **Automated audits:** Software tools ka istemal karke technical aur accessibility ki galtiyen pakadna, jaise code errors ya broken links.
 - **Keyboard and screen-reader walkthroughs:** Sirf assistive technologies (jaise screen reader) use karke check karna ki

disabled log (mazoor afrad) system aasaani se chala sakte hain ya nahi.

- **Prototype testing:** App ke early design (mock-ups) ko control conditions mein users ke sath test karna.
- **A/B testing:** Ek hi app/website ke do (two) versions ko aapas mein compare karna taaki pata chale ki konsa version zyada behtar performance de raha hai.
- **Compatibility testing:** Yeh check karna ki system alag-alag browsers, devices, aur operating systems par theek se chalta hai.
- **Performance and load testing:** System par bohot zyada user traffic (load) daal kar uski speed, stability aur responsiveness check karna.

Topic 12: Career Opportunities (HCI padhne ke baad Jobs)

- **Teacher Key Points:**

- HCI, Usability, aur UI/UX padhne ke baad digital product design ki field mein bachon ke liye bohot se shandaar career options khulte hain.
- Kuch aham jobs yeh hain:
 - **UX Designer:** User ki zarooriyat par research karta hai aur product ko aasan (intuitive) aur satisfying banata hai.
 - **UI Designer:** Product ka visual layout (jaise buttons, menus, typography) design karta hai.
 - **Interaction Designer:** Is baat par focus karta hai ki user system ke sath touch, voice ya gesture ke zariye kaise interact karega.
 - **Accessibility Specialist:** Yeh make sure karta hai ki app ko har tarah ke log (jaise disabled log) aasaani se use kar sakein.
 - **Usability Researcher/Tester:** Asli users ke sath testing (jaise SUS ya NPS) karta hai taaki app launch hone se pehle aur baad mein behtar ho sake.
 - **Product Designer:** UX, UI aur business understanding ko mila kar pura end-to-end digital product design karta hai.

1. Encircle the correct answer (MCQs ke Sahi Jawab):

- i) What is the main goal of HCI? **(c) Make systems user-friendly.**
- ii) Which interaction type uses voice commands? **(b) Voice interaction.**
- iii) Which element of UI design includes buttons, menus, and layouts? **(b) Interface Design.**
- iv) What is wireframing used for? **(c) Designing layout.**
- v) What is the role of a prototype in HCI design? **(b) Early interactive model for testing ideas.**
- vi) What happens when feedback in an interface is weak? **(b) Users feel confused and uncertain.**
- vii) Which factor makes systems harder to use? **(c) Inconsistent design.**
- viii) What is the effect of high cognitive load in HCI? **(b) Users become frustrated and make errors.**
- ix) What is the main goal of accessibility testing? **(c) To verify usability for people with disabilities.**
- x) Why is usability testing important in HCI? **(b) To identify problems before release.**

2. Fill in the Blanks (Sahi Alfaz ka Istemal):

- i) HCI is important because it transforms **complex** systems into everyday tools that are usable and enjoyable.
- ii) In traditional interaction, users rely on devices like **mouse** and **keyboard** to communicate with computers.
- iii) Accessibility principles such as **symbols** and **labels** ensure inclusivity for colour-blind users.
- iv) A prototype is an early **interactive** model used to test design ideas before final development.
- v) UI refers to the visual layout of buttons, menus, and screens, while UX refers to the overall **experience** of using the system.

